EXPERIMENT – 22

Prolog Program to Check Whether a Particular Bird Can Fly or Not

Aim:

To write and execute a Prolog program to check whether a particular bird can fly or not using facts and rules, incorporating the required queries.

Program

File Name: exp22.pl

File Path: C:/ai prolog/exp22.pl

```
% Facts about birds

bird(sparrow).
bird(parrot).
bird(eagle).
bird(pigeon).
bird(ostrich).
bird(penguin).
bird(peacock).

% Birds that can fly

can_fly(sparrow).
can_fly(parrot).
can_fly(eagle).
can_fly(pigeon).
can_fly(peacock).

% Birds that cannot fly
```

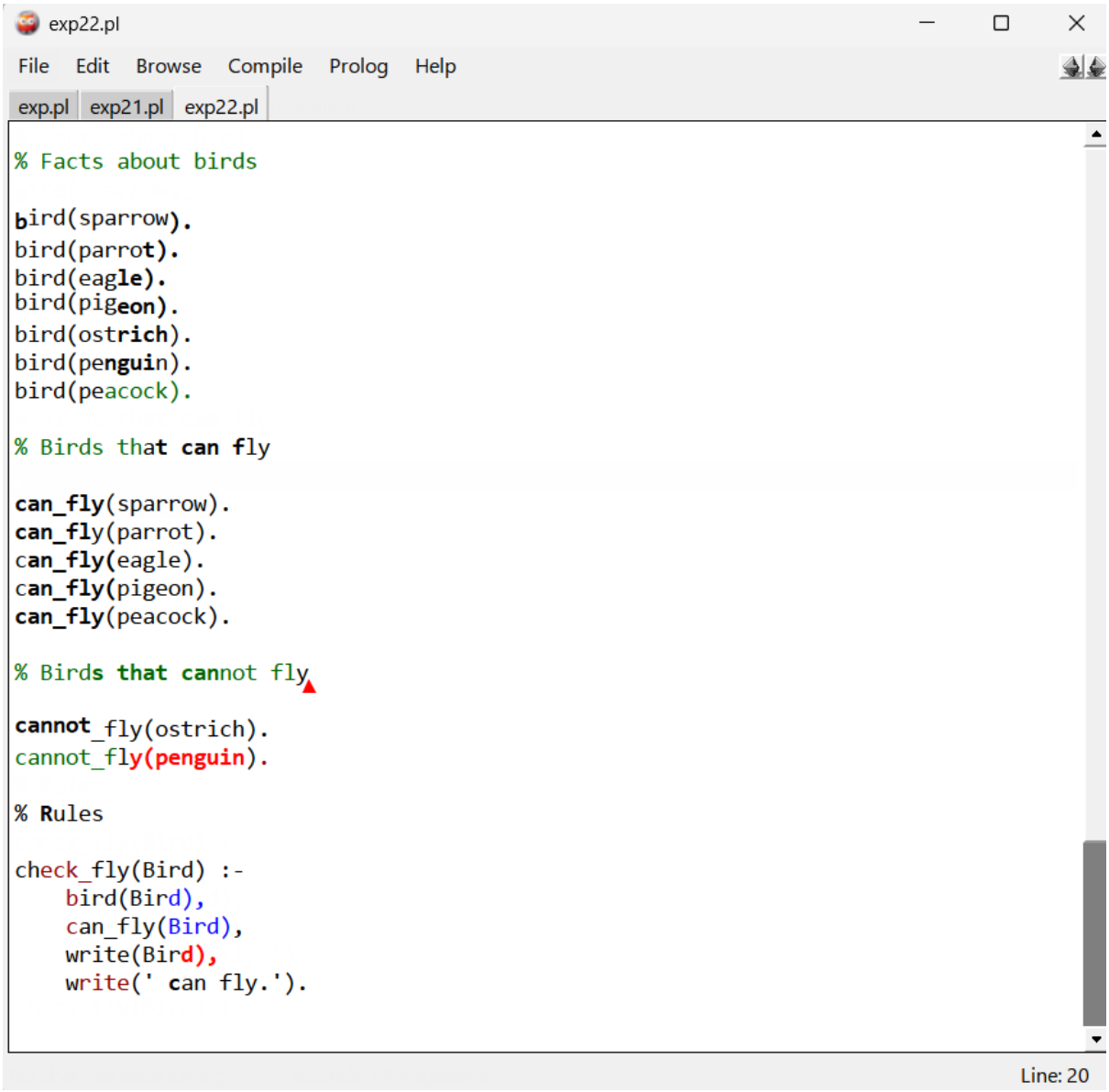
```
cannot_fly(ostrich).
cannot_fly(penguin).

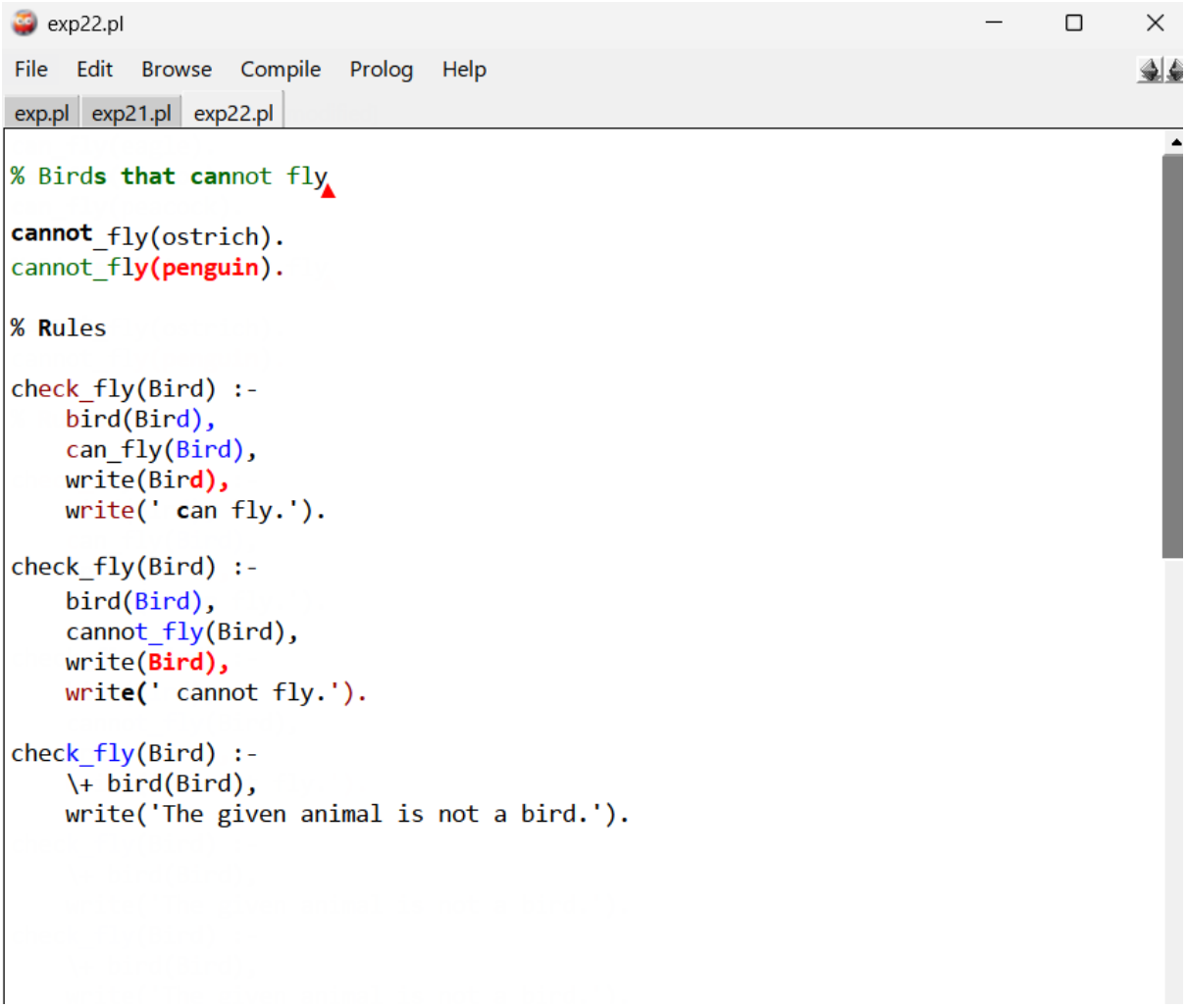
% Rules

check_fly(Bird) :-
   bird(Bird),
   can_fly(Bird),
   write(Bird),
   write(' can fly.').

check_fly(Bird) :-
   bird(Bird),
   cannot_fly(Bird),
   write(Bird),
   write(' cannot fly.').

check_fly(Bird) :-
   \+ bird(Bird),
   write('The given animal is not a bird.').
```


exp22.pl
File Edit Browse Compile Prolog Help
exp.pl exp21.pl exp22.pl
% Facts about birds
bird(sparrow).
bird(parrot).
bird(eagle).
bird(pigeon).
bird(ostrich).
bird(penguin).
bird(peacock).
% Birds that can fly
can_fly(sparrow).
can_fly(parrot).
can_fly(eagle).
can_fly(pigeon).
can_fly(peacock).
% Birds that cannot fly
cannot_fly(ostrich).
cannot_fly(penguin).
% Rules
check_fly(Bird) :-
bird(Bird),
can_fly(Bird),
write(Bird),
write(' can fly.').
Line: 20



Procedure

1. **Open SWI-Prolog software.**
2. **Create a new file and type the above program.**
3. **Save the file as exp22.pl in C:/ai prolog/.**
4. **Load the program using the following command:**

?- consult('C:/ai prolog/exp22.pl').

5. **Enter the required queries to check whether a particular bird can fly or not.**
6. **Observe the output in the Prolog console.**

Required Queries and Output

Query 1: Check whether a sparrow can fly.

?- check_fly(sparrow).

Output:

```
42 ?- check_fly(sparrow).
sparrow can fly.
true
  Possible actions:
  ; (n,r,space,TAB): redo               | t:           trace&redo
  *:                 show choicepoint   | . (c,a,RET): stop
  w:                 write              | p:           print
  +:                 max_depth*5        | -:           max_depth//5
  b:                 break              | h (?):       help

Action? [max_depth(2)]
true ;
false.

43 ?- check_fly(penguin).
penguin cannot fly.
true .

44 ?-
```

Explanation

- **Facts: Define the birds and their flying abilities.**
- **Rules: Determine whether a given bird can fly or cannot fly.**
- **Queries: Test the flying ability of a particular bird.**
- **The \+ operator represents negation as failure in Prolog.**

Result

Thus, the Prolog program to identify whether a particular bird can fly or not was written and executed successfully using facts, rules, and queries.